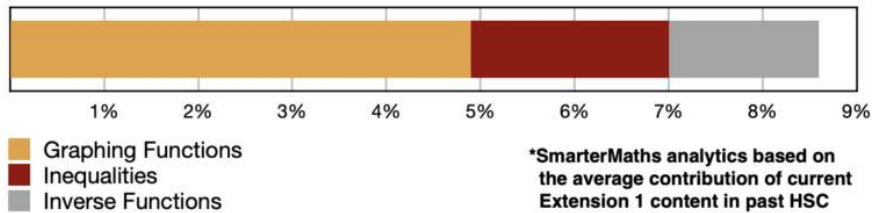


EXT 1: Functions (Ext1), F1 Further Work With Functions (Y11) Inequalities (Ext1)

Teacher: Ljiljana Todoc

Exam Equivalent Time: 115.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

F1 Functions



*SmarterMaths analytics based on the average contribution of current Extension 1 content in past HSC exams over the last decade.

HISTORICAL CONTRIBUTION

- *Functions* is expected to be a significant contributor to the Ext1 exam (our analysis suggests ~9%) and is the combination of old Ext2, Ext1 and Advanced course materials, plus new content!
- This topic has been split into four categories for analysis purposes which are: 1-*Graphing* (4.9%), 2-*Inequalities* (2.1%), 3-*Inverse Functions* (1.6%) and 4-*Parametric Equations* (new content).
- This analysis looks at *Inequalities*.

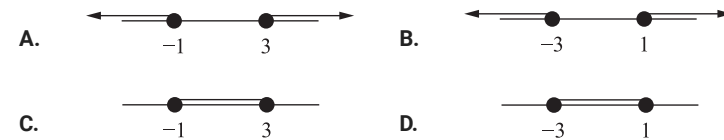
HSC ANALYSIS - What to expect and common pitfalls

- *Inequalities* (2.1%) has been allocated at least one dedicated question in the Ext1 exam every year since 2009, with the exception of 2018.
- When examined, the mark allocation has been a minimum of 3 marks, with 2020 and 2013 the only outliers with an allocation of just 1 mark.
- Algebraic fraction inequalities are easily the most common question type from past Ext1 papers although it will now have to compete with recategorised *Mathematics* course content of absolute value inequalities and regions.
- Absolute value inequalities were regularly examined in the old *Mathematics* course and are a present from examiners if asked. Numerous examples are included in the database.
- Graphing inequality regions was a sneaky contributor to the old *Mathematics* course, asked 6 times in the last decade and producing sub-50% mean marks half the time. Numerous examples can be found in the database.
- The content in this area is expected to present band 3-4 level difficulty questions with meaningful mark allocations. Revision here should aim of eliminate errors and get ahead of the exam clock, *in that order!*

Questions

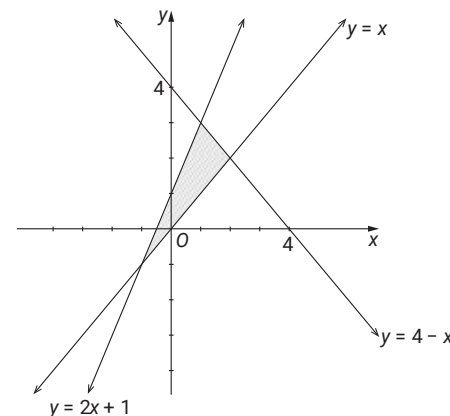
1. Functions, EXT1 F1 2020 HSC 1 MC

Which diagram best represents the solution set of $x^2 - 2x - 3 \geq 0$?



2. Functions, EXT1* F1 2017 HSC 8 MC

The region enclosed by $y = 4 - x$, $y = x$ and $y = 2x + 1$ is shaded in the diagram below.



Which of the following defines the shaded region?

- (A) $y \leq 2x + 1$, $y \leq 4 - x$, $y \geq x$
- (B) $y \geq 2x + 1$, $y \leq 4 - x$, $y \geq x$
- (C) $y \leq 2x + 1$, $y \geq 4 - x$, $y \geq x$
- (D) $y \geq 2x + 1$, $y \geq 4 - x$, $y \geq x$

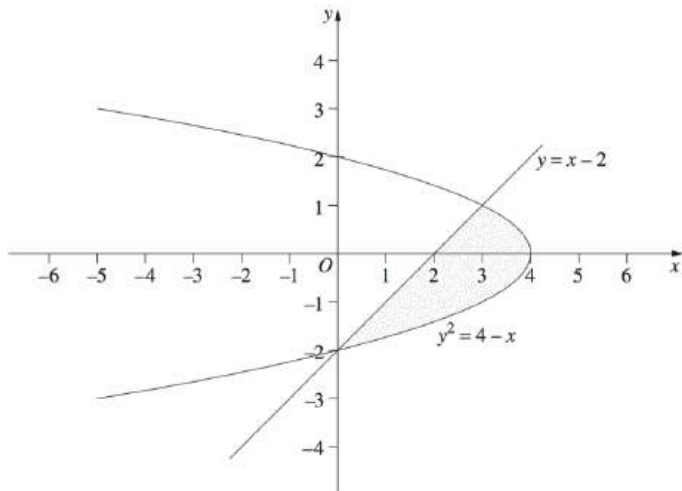
3. Functions, EXT1 F1 2013 HSC 10 MC

Which inequality has the same solution as $|x + 2| + |x - 3| = 5$?

- (A) $\frac{5}{3-x} \geq 1$
 (B) $\frac{1}{x-3} - \frac{1}{x+2} \leq 0$
 (C) $x^2 - x - 6 \leq 0$
 (D) $|2x - 1| \geq 5$

4. Functions, EXT1* F1 2012 HSC 8 MC

The diagram shows the region enclosed by $y = x - 2$ and $y^2 = 4 - x$.



Which of the following pairs of inequalities describes the shaded region in the diagram?

- (A) $y^2 \leq 4 - x$ and $y \leq x - 2$
 (B) $y^2 \leq 4 - x$ and $y \geq x - 2$
 (C) $y^2 \geq 4 - x$ and $y \leq x - 2$
 (D) $y^2 \geq 4 - x$ and $y \geq x - 2$

5. Functions, EXT1* F1 2004 HSC 1f

Find the values of x for which $|x + 1| \leq 5$. (2 marks)

6. Functions, EXT1* 2005 HSC 1e

Find the values of x for which $|x - 3| \leq 1$. (2 marks)

7. Functions, EXT1* F1 2012 HSC 11b

Solve $|3x - 1| < 2$ (2 marks)

8. Functions, EXT1 F1 2004 HSC 1b

Solve $\frac{4}{x+1} < 3$. (3 marks)

9. Functions, EXT1 F1 2010 HSC 1d

Solve $\frac{3}{x+2} < 4$. (3 marks)

10. Functions, EXT1 F1 2011 HSC 1c

Solve $\frac{4-x}{x} < 1$. (3 marks)

11. Functions, EXT1 F1 2012 HSC 11c

Solve $\frac{x}{x-3} < 2$. (3 marks)

12. Functions, EXT1 F1 2014 HSC 11e

Solve $\frac{x^2 + 5}{x} > 6$. (3 marks)

13. Functions, EXT1 F1 2019 HSC 11b

For what values of x is $\frac{x}{x+1} < 2$? (3 marks)

14. Functions, EXT1 F1 SM-Bank 1

Find the values of x that satisfy the equation

$$x^2 + 8x + 3 \leq 0. \text{ (3 marks)}$$

15. Functions, EXT1 F1 SM-Bank 2

Solve $\frac{3}{|x-3|} < 3$. (3 marks)

16. Functions, EXT1* F1 2010 HSC 2b

Solve the inequality $x^2 - x - 12 < 0$. (2 marks)

17. Functions, EXT1* F1 2016 HSC 11c

Solve $|x-2| \leq 3$. (2 marks)

18. Functions, EXT1 F1 2004 HSC 1a

Indicate the region on the number plane satisfied by $y \geq |x+1|$. (2 marks)

19. Functions, EXT1 F1 2015 HSC 11c

Solve the inequality $\frac{4}{x+3} \geq 1$. (3 marks)

20. Functions, EXT1 F1 2017 HSC 11c

Solve $\frac{2x}{x+1} > 1$. (3 marks)

21. Functions, EXT1 F1 SM-Bank 3

A circle has centre $(5, 3)$ and radius 3.

- Describe, with inequalities, the region that consists of the interior of the circle and more than 2 units above the x -axis. (2 marks)
- Sketch the region. (1 mark)

22. Functions, EXT1 F1 2008 HSC 3a

- Sketch the graph of $y = |2x-1|$. (1 mark)
- Hence, or otherwise, solve $|2x-1| \leq |x-3|$. (3 marks)

23. Functions, EXT1 F1 2009 HSC 1d

Solve the inequality $\frac{x+3}{2x} > 1$. (3 marks)

24. Functions, EXT1 F1 2016 HSC 11e

Solve $\frac{3}{2x+5} - x > 0$. (3 marks)

25. Functions, EXT1* F2 2019 HSC 11g

The parabola $y = x^2$ meets the line $y = x+2$ at the points $(-1, 1)$ and $(2, 4)$. Do NOT prove this.

By first sketching the graphs of $y = x^2$ and $y = x+2$, shade the region which simultaneously satisfies the two inequalities $y \geq x^2$ and $y \geq x+2$. (2 marks)

26. Functions, EXT1 F1 2007 HSC 3b

i. Find the vertical and horizontal asymptotes of the hyperbola $y = \frac{x-2}{x-4}$

and hence sketch the graph of $y = \frac{x-2}{x-4}$. (3 marks)

ii. Hence, or otherwise, find the values of x for which

$$\frac{x-2}{x-4} \leq 3. \quad (2 \text{ marks})$$

27. Functions, EXT1* F1 2009 HSC 3c

Shade the region in the plane defined by $y \geq 0$ and $y \leq 4 - x^2$. (2 marks)

28. Functions, EXT1* F1 2013 HSC 11g

Sketch the region defined by $(x-2)^2 + (y-3)^2 \geq 4$. (3 marks)

29. Functions, EXT1* F2 2015 HSC 13b

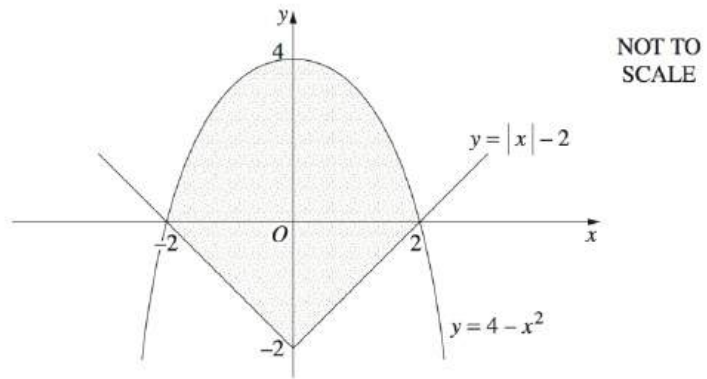
i. Find the domain and range for the function $f(x) = \sqrt{9-x^2}$. (2 marks)

ii. On a number plane, shade the region where the points (x, y) satisfy both of the inequalities

$$y \leq \sqrt{9-x^2} \text{ and } y \geq x. \quad (2 \text{ marks})$$

30. Functions, EXT1* F2 2011 HSC 4e

The diagram shows the graphs $y = |x| - 2$ and $y = 4 - x^2$.



Write down inequalities that together describe the shaded region. (2 marks)

Worked Solutions

1. Functions, EXT1 F1 2020 HSC 1 MC

$$x^2 - 2x - 3 \geq 0$$

$$(x - 3)(x + 1) \geq 0$$

$$x \geq 3, \text{ or } x \leq -1$$

$$\Rightarrow A$$

2. Functions, EXT1* F1 2017 HSC 8 MC

Consider $y = 2x + 1$,

Shading is below graph

$$\Rightarrow y \leq 2x + 1$$

Consider $y = 4 - x$,

Shading is below graph

$$\Rightarrow y \leq 4 - x$$

$$\Rightarrow A$$

3. Functions, EXT1 F1 2013 HSC 10 MC

In A and B , $x \neq 3$ but when $x = 3$,
 $|3 + 2| + |3 - 3| = 5$ is correct.
 $\therefore$ Not A or B .

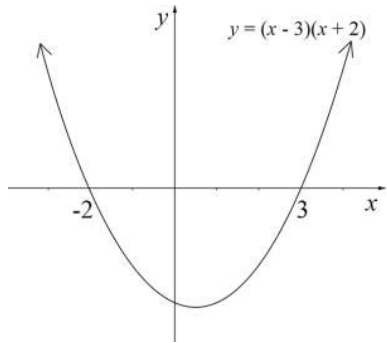
Consider D

$x \rightarrow \infty$ satisfies $|2x - 1| \geq 5$, but
 obviously not $|x + 2| + |x - 3| = 5$.

Consider C

$$x^2 - x - 6 \leq 0$$

$$(x - 3)(x + 2) \leq 0$$



True when $-2 \leq x \leq 3$.

In this range,

$$(x + 2) \geq 0 \text{ and } (x - 3) \leq 0$$

$\therefore$ We can write

$$|x + 2| + |x - 3| = (x + 2) - (x - 3)$$

$$= x + 2 - x + 3$$

$$= 5$$

$\therefore C$ has the same solution

$\Rightarrow C$ is correct.

♦♦ Mean mark 39%
COMMENT: Note that the quick
 elimination of A , B and D is
 sufficient to get to the correct
 answer without proving C
 (although we have done this in the
 solution).

4. Functions, EXT1* F1 2012 HSC 8 MC

Using information from diagram

$(3, 0)$ is in the shaded region

Substituting $(3, 0)$ into $y^2 \leq 4 - x$, $0 \leq 4 - 3 \Rightarrow \text{true}$

$\therefore$ Cannot be C or D

Similarly

$(3, 0)$ must satisfy other inequality

i.e. $y \leq x - 2$ becomes $0 \leq 3 - 2 \Rightarrow \text{true}$

$\Rightarrow A$

♦ Mean mark 44%.

5. Functions, EXT1* F1 2004 HSC 1f

Solution 1

$$|x + 1| \leq 5$$

$$-5 \leq x + 1 \leq 5$$

$$\therefore -6 \leq x \leq 4$$

Solution 2

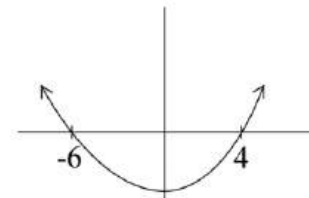
$$|x + 1| \leq 5$$

$$(x + 1)^2 \leq 5^2$$

$$x^2 + 2x + 1 \leq 25$$

$$x^2 + 2x - 24 \leq 0$$

$$(x + 6)(x - 4) \leq 0$$



$$\therefore -6 \leq x \leq 4$$

6. Functions, EXT1* 2005 HSC 1e

$$|x - 3| \leq 1$$

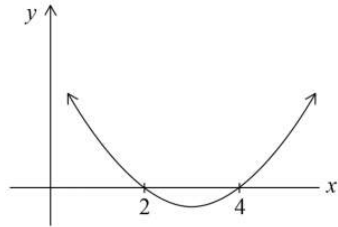
Solution 1

$$(x - 3)^2 \leq 1$$

$$x^2 - 6x + 9 \leq 1$$

$$x^2 - 6x + 8 \leq 0$$

$$(x - 4)(x - 2) \leq 0$$



$$\therefore 2 \leq x \leq 4$$

Alternative Solution

$$(x - 3) \leq 1 \quad -(x - 3) \leq 1$$

$$x \leq 4 \quad -x + 3 \leq 1$$

$$-x \leq -2$$

$$x \geq 2$$

$$\therefore 2 \leq x \leq 4$$

7. Functions, EXT1* F1 2012 HSC 11b

$$|3x - 1| < 2$$

$$3x - 1 < 2 \quad -(3x - 1) < 2$$

$$3x < 3 \quad -3x + 1 < 2$$

$$x < 1 \quad 3x > -1$$

$$x > -\frac{1}{3}$$

$$\therefore -\frac{1}{3} < x < 1$$

MARKER'S COMMENT: Note that both conditions must be satisfied! Dealing with negative signs and division for inequalities produced many errors.

8. Functions, EXT1 F1 2004 HSC 1b

$$\frac{4}{x+1} < 3$$

Multiply both sides by $(x + 1)^2$

$$4(x + 1) < 3(x + 1)^2$$

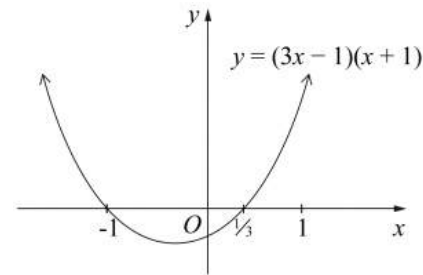
$$4x + 4 < 3(x^2 + 2x + 1)$$

$$4x + 4 < 3x^2 + 6x + 3$$

$$3x^2 + 2x - 1 > 0$$

$$(3x - 1)(x + 1) > 0$$

$$\text{LHS} = 0 \text{ when } x = \frac{1}{3} \text{ or } -1$$



From the graph,

$$x < -1 \text{ or } x > \frac{1}{3}$$

9. Functions, EXT1 F1 2010 HSC 1d

Solution 1

$$\frac{3}{x+2} < 4$$

Multiply b.s. by $(x+2)^2$

$$3(x+2) < 4(x+2)^2$$

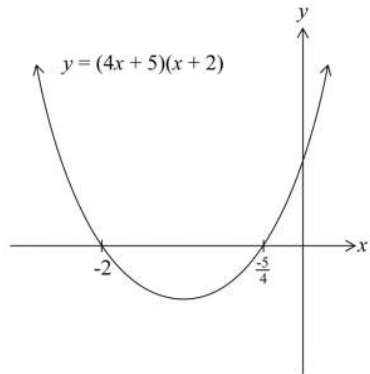
$$3x+6 < 4(x^2+4x+4)$$

$$3x+6 < 4x^2+16x+16$$

$$4x^2+13x+10 > 0$$

$$(4x+5)(x+2) > 0$$

LHS = 0 when $x = -\frac{5}{4}$ or -2



From graph

$$x < -2 \text{ or } x > -\frac{5}{4}$$

Alternate Solution

If $x+2 > 0$ (i.e. $x > -2$)

$$3 < 4(x+2)$$

$$3 < 4x+8$$

$$4x > -5$$

$$x > -\frac{5}{4}$$

If $x+2 < 0$ (i.e. $x < -2$)

$$3 > 4(x+2)$$

$$3 > 4x+8$$

$$4x < -5$$

$$x < -\frac{5}{4}$$

$\therefore x < -2$ (satisfies both)

$\therefore x < -2$ or $x > -\frac{5}{4}$

10. Functions, EXT1 F1 2011 HSC 1c

Solution 1

$$\frac{4-x}{x} < 1$$

$$\text{If } x < 0, \quad 4 - x > x$$

$$2x < 4$$

$$x < 2$$

$$\Rightarrow x < 0 \text{ (satisfies both)}$$

$$\text{If } x > 0, \quad 4 - x < x$$

$$2x > 4$$

$$x > 2$$

$$\Rightarrow x > 2 \text{ (satisfies both)}$$

$$\therefore x < 0 \text{ or } x > 2$$

Solution 2

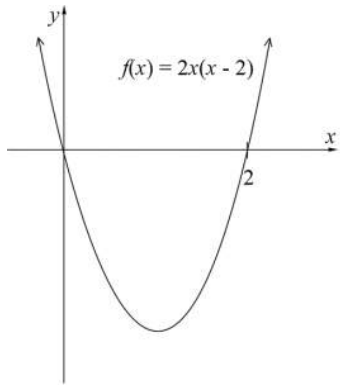
Multiply both sides by x^2

$$x(4-x) < x^2$$

$$4x - x^2 < x^2$$

$$2x^2 - 4x > 0$$

$$2x(x-2) > 0$$



From graph

$$x < 0 \text{ or } x > 2$$

11. Functions, EXT1 F1 2012 HSC 11c

Solution 1

$$\frac{x}{x-3} < 2$$

When $x - 3 > 0$, (i.e. $x > 3$)

$$x < 2(x - 3)$$

$$x < 2x - 6$$

$$\Rightarrow x > 6 \text{ (satisfies both)}$$

When $x - 3 < 0$, (i.e. $x < 3$)

$$x > 2(x - 3)$$

$$x > 2x - 6$$

$$x < 6$$

$$\Rightarrow x < 3 \text{ (satisfies both)}$$

$\therefore$ Equation is correct for $x < 3$ or $x > 6$

Solution 2

$$\frac{x}{(x-3)} < 2$$

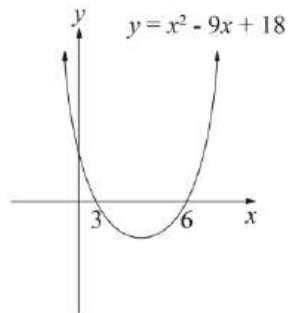
Multiply b.s. by $(x-3)^2$

$$x(x-3) < 2(x^2 - 6x + 9)$$

$$x^2 - 3x < 2x^2 - 12x + 18$$

$$x^2 - 9x + 18 > 0$$

$$(x-3)(x-6) > 0$$



$\therefore$ Equation correct for $x < 3$ or $x > 6$

12. Functions, EXT1 F1 2014 HSC 11e

$$\frac{x^2 + 5}{x} > 6$$

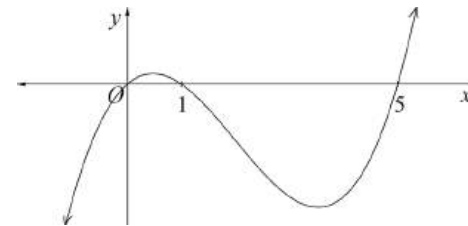
$$x^2 \left(\frac{x^2 + 5}{x} \right) > 6x^2$$

$$x^3 + 5x > 6x^2$$

$$x^3 - 6x^2 + 5x > 0$$

$$x(x^2 - 6x + 5) > 0$$

$$x(x-5)(x-1) > 0$$



$\therefore 0 < x < 1$ or $x > 5$

13. Functions, EXT1 F1 2019 HSC 11b

$$\frac{x}{x+1} < 2$$

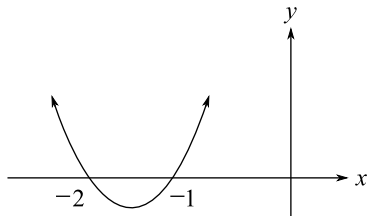
Multiply b.s. by $(x+1)^2$

$$x(x+1) < 2(x+1)^2$$

$$x^2 + x < 2x^2 + 4x + 2$$

$$0 < x^2 + 3x + 2$$

$$0 < (x+2)(x+1)$$



From graph:

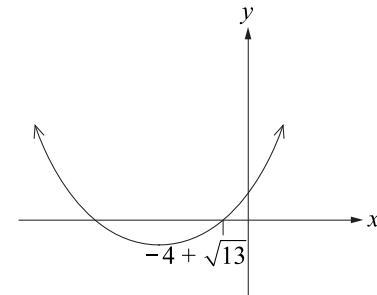
$$x < -2 \text{ or } x > -1$$

14. Functions, EXT1 F1 SM-Bank 1

$$x = \frac{-8 \pm \sqrt{8^2 - 4 \cdot 1 \cdot 3}}{2}$$

$$= \frac{-8 \pm \sqrt{52}}{2}$$

$$= -4 \pm \sqrt{13}$$



$$\{x: x \leq -4 - \sqrt{3} \cup x \geq -4 + \sqrt{3}\}$$

15. Functions, EXT1 F1 SM-Bank 2

Solution 1

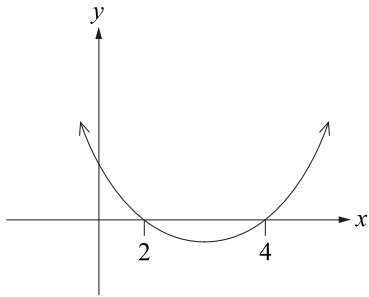
$$\frac{3}{|x-3|} < 3$$

$$|x-3| > 1$$

$$(x^2 - 6x + 9) > 1^2$$

$$x^2 - 6x + 8 > 0$$

$$(x-4)(x-2) > 0$$



$$\therefore \{x: x < 2 \cup x > 4\}$$

Solution 2

$$|x-3| > 1$$

If $(x-3) > 0$, i.e. $x > 3$

$$x-3 > 1$$

$$x > 4$$

$$\Rightarrow x > 4 \text{ (satisfies both)}$$

If $(x-3) < 0$, i.e. $x < 3$

$$-(x-3) > 1$$

$$-x+3 > 1$$

$$x < 2$$

$$\Rightarrow x < 2 \text{ (satisfies both)}$$

$$\therefore \{x: x < 2 \cup x > 4\}$$

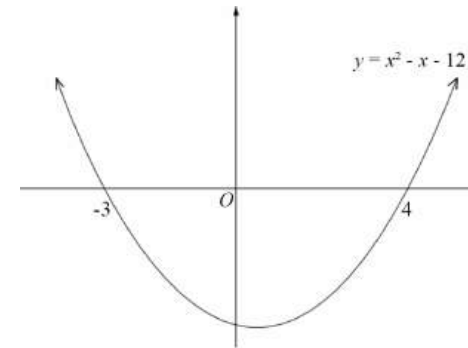
16. Functions, EXT1* F1 2010 HSC 2b

$$x^2 - x - 12 < 0$$

$$\text{Solve } x^2 - x - 12 = 0$$

$$(x-4)(x+3) = 0$$

$$\Rightarrow x = 4 \text{ or } -3$$



$$x^2 - x - 12 < 0$$

$$\text{when } 3 < x < 4$$

MARKER'S COMMENT: Drawing the parabola proved the most efficient way of answering this question for students.

17. Functions, EXT1* F1 2016 HSC 11c

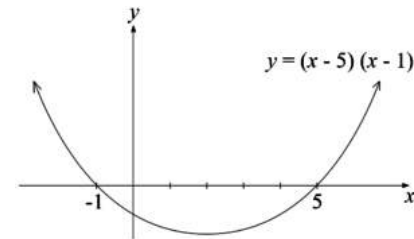
$$|x-2| \leq 3$$

$$(x-2)^2 \leq 3^2$$

$$(x^2 - 4x + 4) \leq 9$$

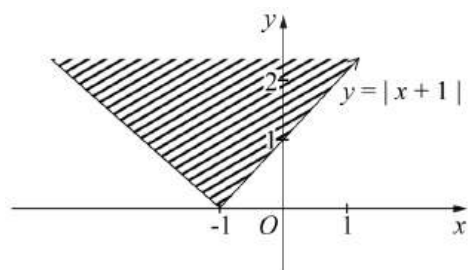
$$x^2 - 4x - 5 \leq 0$$

$$(x-5)(x+1) \leq 0$$



$$\therefore -1 \leq x \leq 5$$

18. Functions, EXT1 F1 2004 HSC 1a



$$y \geq |x + 1|$$

Test (0, 0)

$$0 \geq |0 + 1|$$

$$0 \geq 1 \Rightarrow \text{False}$$

$\therefore$ Shaded area represents

$$y \geq |x + 1|$$

19. Functions, EXT1 F1 2015 HSC 11c

Solution 1

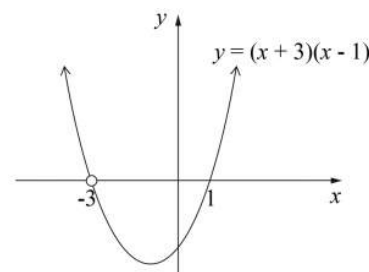
$$\frac{4}{x + 3} \geq 1$$

$$4(x + 3) \geq (x + 3)^2$$

$$4x + 12 \geq x^2 + 6x + 9$$

$$x^2 + 2x - 3 \leq 0$$

$$(x + 3)(x - 1) \leq 0$$



$$\therefore -3 < x \leq 1, x \neq -3$$

Solution 2

$$\text{If } (x + 3) > 0$$

$$x > -3$$

$$\frac{4}{x + 3} \geq 1$$

$$4 \geq x + 3$$

$$x \leq 1$$

$$\therefore -3 < x \leq 1$$

$$\text{If } (x + 3) < 0$$

$$x < -3$$

$$\frac{4}{x + 3} \geq 1$$

$$4 \leq x + 3$$

$$x \geq 1$$

$\therefore$ No solution.

$$\therefore -3 < x \leq 1$$

20. Functions, EXT1 F1 2017 HSC 11c

$$\frac{2x}{x+1} > 1$$

If $x+1 > 0$, i.e. $x > -1$

$$2x > x+1$$

$$x > 1$$

$$\Rightarrow x > 1$$

If $x+1 < 0$, i.e. $x < -1$

$$2x < x+1$$

$$x < 1$$

$$\Rightarrow x < -1$$

$$\therefore x < -1 \text{ or } x > 1$$

21. Functions, EXT1 F1 SM-Bank 3

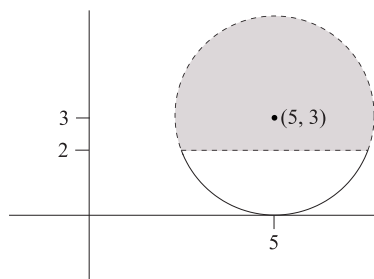
i. Equation of circle:

$$(x-5)^2 + (y-3)^2 = 3^2$$

$\therefore$ Region is:

$$(x-5)^2 + (y-3)^2 < 9 \cap y > 2$$

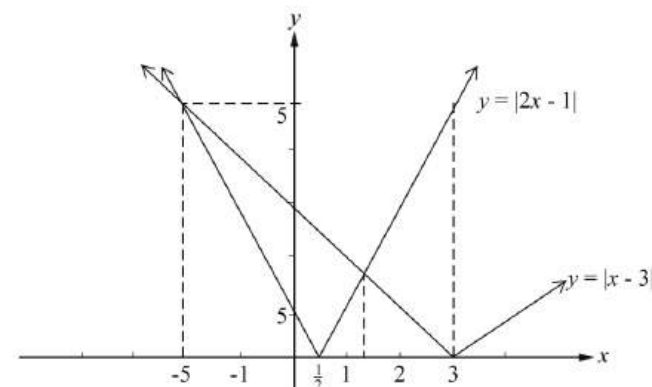
ii.



COMMENT: The broken line on the graph represents an excluded boundary.

22. Functions, EXT1 F1 2008 HSC 3a

i.



ii. Solving for $|2x - 1| \leq |x - 3|$

Graph shows the statement is TRUE

between the points of intersection.

$\Rightarrow$ Intersection occurs when

$$(2x - 1) = (x - 3) \text{ or } -(2x - 1) = x - 3$$

$$x = -2$$

$$-2x + 1 = x - 3$$

$$-3x = -4$$

$$x = \frac{4}{3}$$

$$\therefore \text{Solution is } \left\{ x: -2 \leq x \leq \frac{4}{3} \right\}$$

23. Functions, EXT1 F1 2009 HSC 1d

Solution 1

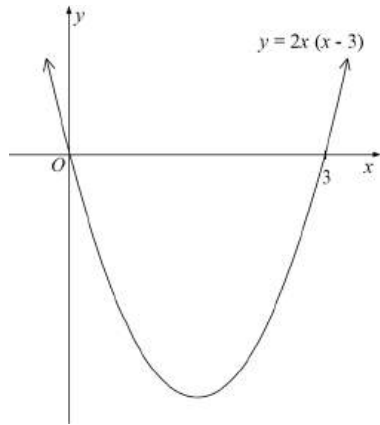
$$(2x)^2 \times \frac{x+3}{2x} > (2x)^2$$

$$2x(x+3) > 4x^2$$

$$2x^2 + 6x > 4x^2$$

$$2x^2 - 6x < 0$$

$$2x(x-3) < 0$$



$$\therefore \{x: 0 < x < 3\}$$

Solution 2

If $x > 0$,

$$x+3 > 2x$$

$$x < 3$$

$$\therefore 0 < x < 3$$

If $x < 0$,

$$x+3 < 2x$$

$$x > 3 \Rightarrow \text{No solution}$$

$$\therefore \{x: 0 < x < 3\}$$

24. Functions, EXT1 F1 2016 HSC 11e

$$\frac{3}{2x+5} - x > 0$$

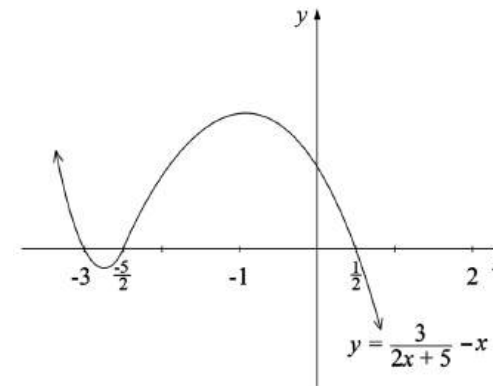
$$3(2x+5) > x(2x+5)^2, \quad x \neq -\frac{5}{2}$$

$$3(2x+5) - x(2x+5)^2 > 0$$

$$(2x+5)[3 - x(2x+5)] > 0$$

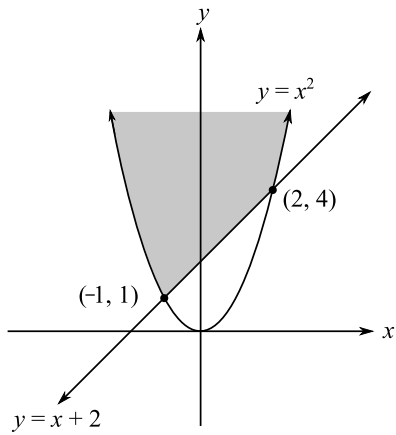
$$(2x+5)(-2x^2 - 5x + 3) > 0$$

$$(2x+5)(1-2x)(x+3) > 0$$



$$x < -3, \quad -\frac{5}{2} < x < \frac{1}{2}$$

25. Functions, EXT1* F2 2019 HSC 11g



26. Functions, EXT1 F1 2007 HSC 3b

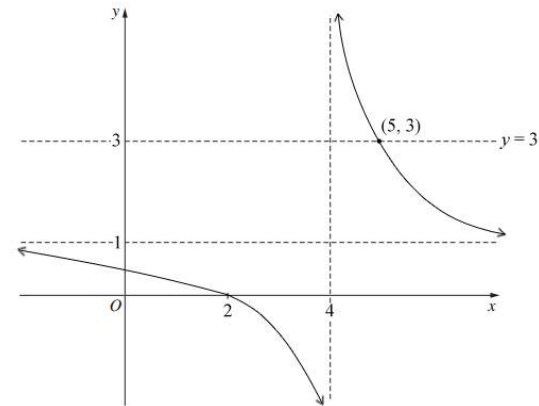
i. $y = \frac{x-2}{x-4}$

Vertical asymptote at $x = 4$

$$\lim_{x \rightarrow \infty} \frac{x-2}{x-4} = \lim_{x \rightarrow \infty} \frac{1 - \frac{2}{x}}{1 - \frac{4}{x}} = 1$$

$$y\text{-intercept} = \frac{1}{2}$$

$$x\text{-intercept} = 2$$



ii. Find x so that $\frac{x-2}{x-4} \leq 3$

$$\text{When } \frac{x-2}{x-4} = 3$$

$$x-2 = 3x-12$$

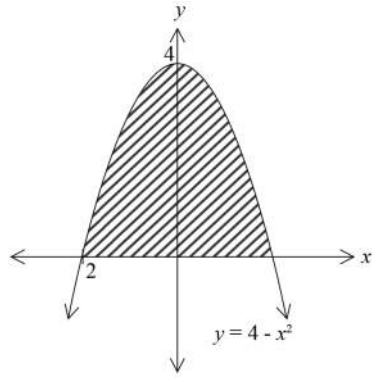
$$2x = 10$$

$$x = 5$$

$$\Rightarrow (5, 3) \text{ is the intersection of } y = 3 \text{ and } y = \frac{x-2}{x-4}$$

$$\therefore \frac{x-2}{x-4} \leq 3 \text{ when } x < 4 \text{ and } x \geq 5.$$

27. Functions, EXT1* F1 2009 HSC 3c

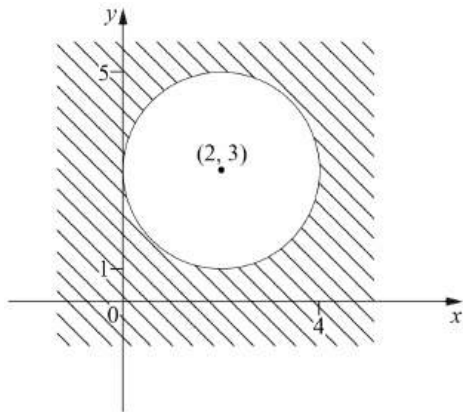


COMMENT: This past "Advanced" HSC question now fits into the Ext1 (new) syllabus.

Shaded area is region where
 $y \geq 0$ and $y \leq 4 - x^2$

28. Functions, EXT1* F1 2013 HSC 11g

The region is the exterior of a circle,
 centre (2,3) and radius 2.



COMMENT: This past "Advanced" HSC question now fits into the Ext1 (new) syllabus.

29. Functions, EXT1* F2 2015 HSC 13b

i. $f(x) = \sqrt{9 - x^2}$

Domain

$$9 - x^2 \geq 0$$

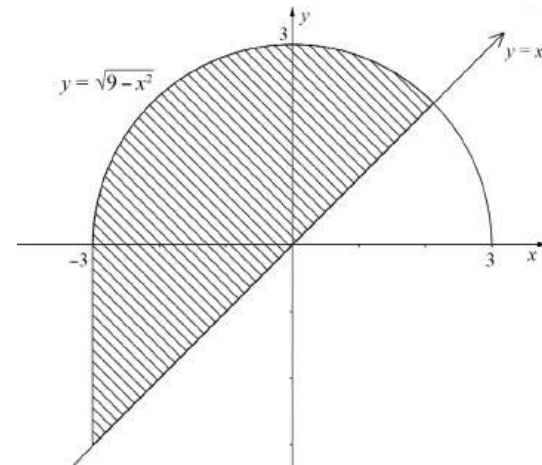
$$x^2 \leq 9$$

$$-3 \leq x \leq 3$$

Range

$$0 \leq y \leq 3$$

ii.



♦ Mean mark 34%.

30. Functions, EXT1* F2 2011 HSC 4e

Inequalities are

$$y \leq 4 - x^2$$

$$y \geq |x| - 2$$

♦ Mean mark 46%.